NTT DATA UK&I
Office of the CTO

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Banking Technology Foresight

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Date

Background: NTT DATA Foresight 2025

Foresight *is NTT DATA's annual report that traces the technology patterns that will become business realities, and recommends strategies for navigating the next wave of change.*

For 2025 our experts have identifed five trends to continue watching. Foresight explores how these trends will impact on organisations and explains how they may evolve over time.

The full Foresight 2025 report can be accessed here.

Foresight 2025 – FS & Banking Sector Context

The "NTT DATA Technology Foresight 2025" report is crucial for the financial services sector as it provides a comprehensive analysis of emerging technological trends and their potential impact on the industry.

The report highlights how advancements in AI, cloud computing, and automation can transform financial services by enhancing decision-making, increasing productivity, and improving customer experiences.

It also emphasizes the importance of AI safety and compliance, data management, and human-machine collaboration, which are essential for leveraging these technologies effectively.

By understanding and adopting these trends, financial institutions can stay competitive, innovate, and meet the evolving needs of their customers in a rapidly changing digital landscape

How to Use Foresight 2025

Foresight 2025 is intended to be a reference for the core concepts NTT DATA forecast will drive a fundamental shift in how we use technology to change the way we do business and how we interact with people and the impact it will have on Society

This supplementary report will translate the underlying Foresight 2025 trends into banking sector points of view to bring relevant and targeted insight.

Foresight 2025: Global, cross-sector trends

These Cross sectoral trends will shape the future of business with the underlying themes having wide ranging impacts. We will explore these themes in the context of banking and consider how we should prepare for this evolution.

Enhanced humans

The synergistic collaboration between people and machines is shaping a future where human potential isn't limited by time, task or knowledge.

Ambient intelligent experiences

Technologies like artificial intelligence (AI), spatial computing and automation are fundamentally changing how organizations connect with their audiences across different touchpoints.

Digital sustainability for economic resilience

A new business strategy is emerging where organizations integrate environmental stewardship with economic growth and assign individual and collective responsibility.

Cognitive cloud convergence

The seamless integration of advanced cloud computing technologies with AI and human cognitive abilities is empowering organizations to improve operations, enhance decision-making and derive deeper insights from their data in real time.

Accelerated security fusion

Capabilities like automated incident response and AI-driven threat detection are enabling organizations to adapt dynamically to emerging threats, ensuring resilience in an ever-changing risk landscape.

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Foresight 2025 – Synergies between Capabilities and Technology

The capability highlighted in this report are complemented by shifts in underlying technology, people and processes. Organisation must manage the impact on each domain to guide adoption of future capabilities.

Future Capabilities

- **Enhanced Decision-Making:** AI provides deeper insights and predictive analytics for better decision-making.
- **Increased Productivity:** Automation of routine tasks allows employees to focus on higher-value activities.
- **Cost Efficiency:** AI reduces costs through better resource allocation and fewer errors.
- **Accelerated Innovation:** AI frees up resources for innovation and development.
- **Improved Customer Experience:** AI enhances customer service with faster and more personalized interactions.

Evolved People, Process and Technology

- **Human-Machine Collaboration:** Fostering effective teamwork between humans and machines, featuring AI-enhanced workplaces with technologies such as chatbots and robotic process automation.
- **Technological Infrastructure and Integration:** Establishing strong cloud systems and specialised hardware for handling large datasets and integrating seamlessly with internal data sources and APIs.
- **Organizational Enablement through AI:** Utilizing AI to improve culture, staff, and processes, and integrating it into HR, supply chain, and financial planning etc.
- **AI Safety and Compliance:** Ensuring AI systems are effective, equitable, legally compliant, and safeguarded against cyber threats and bias while following data protection regulations like GDPR.
- **Data Management and Governance:** Storing and analyzing large amounts of data properly, ensuring quality, privacy, and security adhere to legal requirements.

Foresight 2025: Applying to Banking

Foresight global trends

Enhanced humans

Ambient intelligent experiences

Digital sustainability for economic resilience

Cognitive cloud convergence

Accelerated security fusion

Top themes in banking

Customer Centric AI Driven Banking

Providing frictionless, curated and personalised journeys for bank customers as they interact with all aspects of banking services

Intelligent & Responsible Banking

Harnessing automation and artificial intelligence to deliver more efficient and faster service at lower cost, with increased agility to change.

Digital Banking Modernisation

Evolving digital infrastructure and security to underpin digital service evolution leveraging modernisation of legacy and cloud to drive transformation

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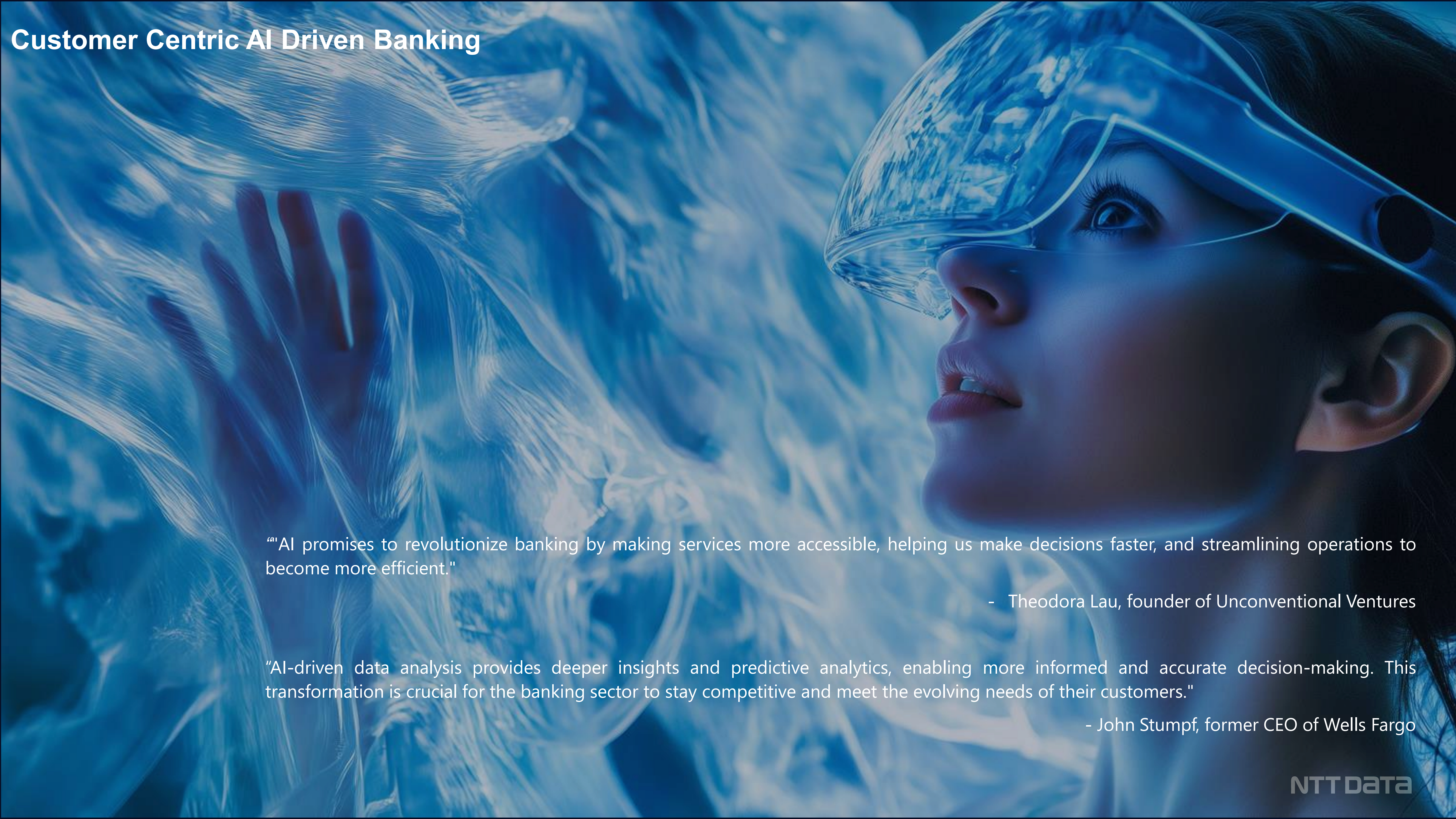


Customer Centric AI Driven Banking

“"AI promises to revolutionize banking by making services more accessible, helping us make decisions faster, and streamlining operations to become more efficient."

- Theodora Lau, founder of Unconventional Ventures

“AI-driven data analysis provides deeper insights and predictive analytics, enabling more informed and accurate decision-making. This transformation is crucial for the banking sector to stay competitive and meet the evolving needs of their customers."

- John Stumpf, former CEO of Wells Fargo

Customer Centric AI Driven Banking : Summany

01 **Rise of Ai Agent banking:** The rise of AI Agents has the potential to transform the banking industry, offering opportunities to improve efficiency, enhance customer and staff experience. This will take the concept of AI driven chatbots to the next level creating and ecosystem of AI enable Agents that can collaborate to deliver better customer outcomes.

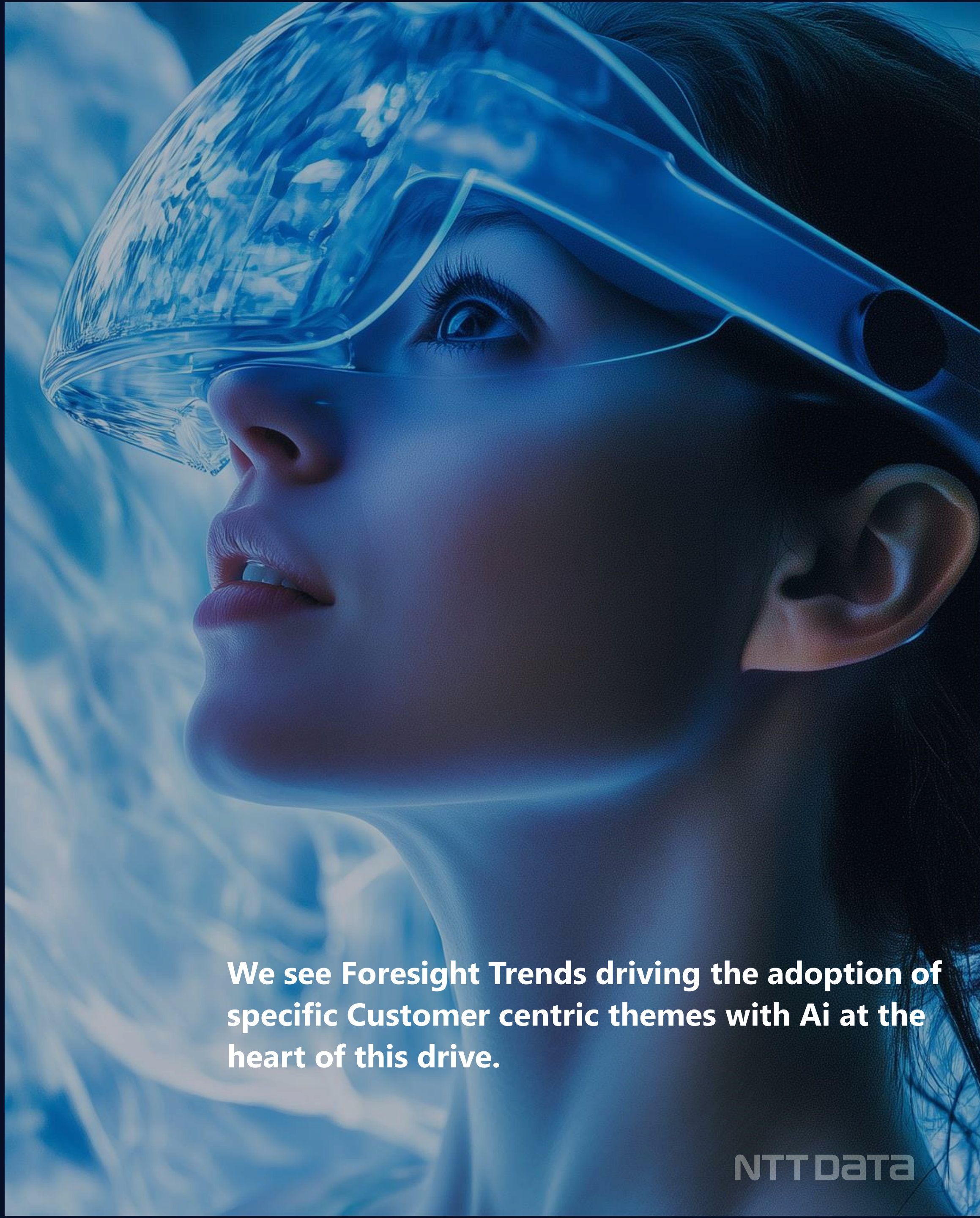
02 **Hyper-personalised Ai-Driven Banking:** Consumers are increasingly demanding personalized experiences in all aspects of their lives, including banking. They want services tailored to their individual needs and preferences. AI-powered systems will analyse customer data to offer personalized financial advice, such as suggesting ways to save money, invest wisely, and manage debt.

03 **Return to Human Relationship:** Digital channels and the closure of physical branches have diminished the human touch in banking, especially in retail and SME sectors. However, advanced technologies can restore this relationship by addressing past cost-efficiency issues.

04 **SME Banking:** Banks are increasingly looking to leverage digital platforms and innovative strategies to enhance the banking experience for small businesses, recognizing them as a significant growth engine in the financial landscape

05 **Augment Workforce:** Banks are modernizing to stay competitive, focusing on both front-end and back-end operations. Manual processes are being replaced by AI. The concept of an augmented workforce is gaining traction, with generative AI transforming the traditional workforce into a more efficient, waste-eliminating "augmented" workforce.

06 **Embedded/Subscription Finance:** Embedded finance and subscription-based banking are converging to offer seamless, integrated financial services within everyday apps, driven by open banking and fintech partnerships. This trend enhances customer engagement and personalization, providing predictable revenue streams for banks



Customer Centric AI Driven Banking : underlying concepts and value

Customer Centric AI Driven Banking

Rise of Ai Agent banking
- Enhanced Convienience
- 24/7 Access
- Targetted & highly personalised
- Authentifcation & Authorisation
- Highly Automated
- Biometrics
- Faster decision making
- Enhanced security and Fraud

Hyper-personalised Ai-Driven Banking
- Tailored Financial Advice
- Proactive Support
- Seamless Omnichannel Experience
- Consumer Trust
- Security
- Product Flexibilty

Human to Human Relationship
- Personalise experiences
- Multi channel support
- Customer Loyalty
- Accessibility
- Social Responsiblity

Digital SME Banking
- Personalise SME banking
- Prompt customer services
- Enhanced Relationship Management
- 24x7 always-on services
- Faster Credit decisioning
- Broad range os SME products and service

Embedded/ Subscriptio n Finance
- High personalised Services
- Value for Money
- Seamless digital Experience
- More Choice
- Integration with non banking eco systems

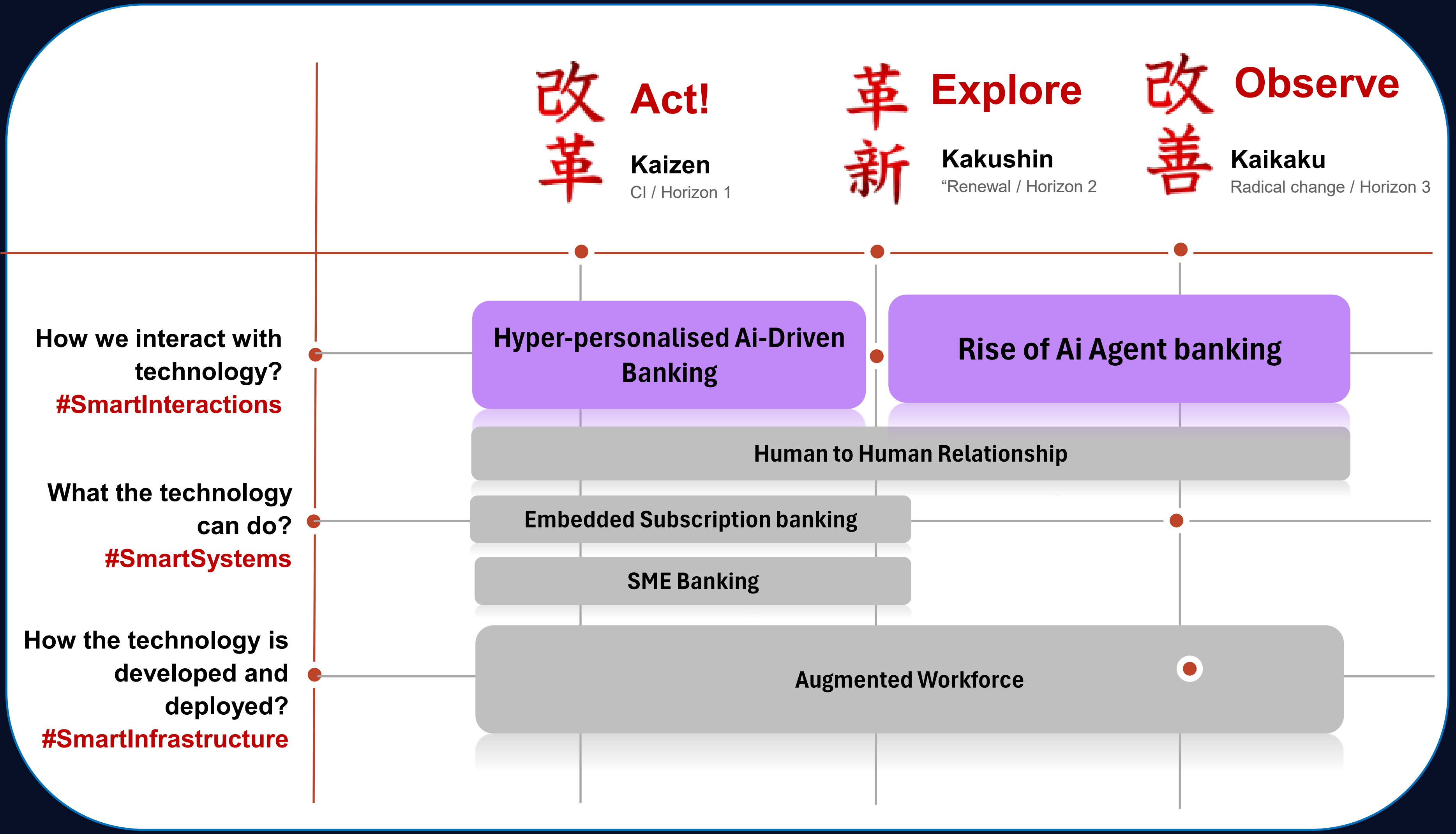
Augmented Workforce
- High personalise sevices
- Alway on enhanced convenienced
- Increased efficiencies through Automation
- Increaced Accuracy
- Better workforce utilisation
- Improce customer satisfaction
- Enhance risk management

Foresight 2025 Technology Horizon : Customer Centric AI Driven Banking

Redefining the relationship between Human and machine

Maintaining the Human Connection throughout while delivering enhanced products and service

Augmented human deliver faster more accurate decisions



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Customer Centric AI Driven Banking

Rise of Ai Agent banking

Rise of Ai Agent banking

- Enhanced Convienience
- 24/7 Access
- Targetted & highly personalise
- Authentifcation & Authorisation
- Highly Automated
- Biometrics
- Faster decision making
- Enhanced security and Fraud

__Overview:__ Bank was an early adopters of RPA and Chatbot technologies with the introduction of Gen AI adding another layer of capability. AI Agents take this to the next level essentially delivering a computer program that can understand, learn and act independently to help with bank tasks, Think of it as a super power Alexa or Google assistant.

They are proactive, can be personalised, always available, automate standard task and can collaborate with other agents. In banking the use cases for Agents are broad but do face compliance and regulatory challenges. Bank need to understand how this technology is evolving to keep up with the trend

Adoption challenges

- Compute, Data Management and integration requirements are complex including, Infra and access to the right Talent to implement
- Maintaining Human touch – Fining the balance between human and AI assistance
- Ensuring responsible AI – managing the risk of bias and compliance for Agents as well as governance of same.

Enablers

- Enhance Strategy for AI adoption to include advanced use cases
- Define Policies for future mode of operating.
- Identify AI technology to invest.

Key Use cases

1. Develop AI Super assistant
2. Real time Financial advisor
3. 24/7 banking model

Key Benefits

1. Better customer experience
2. Improved operational efficiency
3. Improve Fraud and cyber security

Industry examples

<Salesforce> Agent Force AI enabled chatbot agents that can complete task autonomously within guardrails

<DBS Bank> Utilises AI Agents to delivery Hyper Personalise services

01 Enhanced Human

02 Ambient Inteligent Experiences

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Customer Centric AI Driven Banking

Hyper-personalised Ai-Driven Banking

Hyper-personalised Ai-Driven Banking

- Tailored Financial Advice
- Proactive Support
- Seamless Omnichannel Experience
- Consumer Trust
- Security
- Product Flexibilty

Overview: *AI is maturing rapidly: Advancements in AI and machine learning are enabling banks to analyze vast amounts of customer data and provide highly personalized insights and recommendations. Customers expect personalization: Consumers are increasingly demanding personalized experiences in all aspects of their lives, including banking. They want services tailored to their individual needs and preferences. Competition is fierce: Banks are facing competition from fintech companies and Big Tech giants that are already leveraging AI to offer innovative and personalized financial services.*

Adoption challenges	Data Privacy and Security	Consumer Trust	Regulatory compliance and AI risk exposure (EU AI act)	Legacy systems and data silo's
Enablers	Invest in AI and RPA technologies	Ensure Compliance and Responsible AI	Combining AI-powered tools with human expertise to enhance the quality of advice provided	Incremental delivery, pilot environments

01 Enhanced Human

02 Ambient Inteligent Experiences

Key Use cases

1. Banking super app including AI enhanced capabilities and advice with Human support

Key Benefits

1. Better banking experience for customers
2. Automate standard tasks and administration activities
3. Enabled bank staff to make better inform decision faster

Industry examples

Personetics:Their AI platform analyzes customer data to provide insights and recommendations that are tailored to individual needs and goals.

Zingly: They combine generative AI with human agents to provide personalized and efficient customer service.

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Customer Centric AI Driven Banking

Return to Human Relationship

Human to Human Relationship

- Personalise experiences
- Multi channel support
- Customer Loyalty
- Accessibility
- Social Responsiblity

Overview: *Banking has lost its human touch due to the rise of digital channels and the closure of physical branches, particularly in retail and SME sectors. However, cutting-edge technologies can restore this relationship by addressing past cost-efficiency issues. These technologies simplify and personalize access to vast amounts of data instantly and through natural language, making customer interactions more efficient and effective while reducing the demands on employee skills. Additionally, they simplify asset allocation, transforming hundreds of thousands of products into personalized portfolios.*

Adoption challenges	Digital Transformation remove the physical touch point with customers	Maintain the right balance between leveraging technology and maintaining personal connections with customers.	Banks must cater to these diverse needs to provide high-quality customer service across all channels.
Enablers	Unified customer Journeys	Cross functional product teams	Augment Human interaction (Blend of Human and Automated)

Key Use cases

1. Enhanced Customer Service
2. Personalised banking experience
3. Omni Channel Efficiency

Key Benefits

1. Increased Customer loyalty
2. Competitive advantage
3. Enhanced revenue opportunities

Industry examples

Lowell Five Bank – Streamlining New Customer Onboarding

Banco Comafi – 93% ROI with Smart Messaging leveraging WhatsApp

Banorte – Enhancing Digital Banking for Retirees

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Customer Centric AI Driven Banking

Digital SME Banking

SME Banking

- Personalise SME banking
- Prompt customer services
- Enhanced Relationship Management
- 24x7 always-on services
- Faster Credit decisioning
- Broad range os SME products and service

Overview: *SMEs often fall between retail and corporate banking segments, prompting banks to develop tailored solutions for their unique needs. By leveraging digital platforms and innovative strategies, banks aim to enhance the banking experience for small businesses, recognizing them as a significant growth engine in the financial landscape. This shift towards more effective service for SME and corporate clients reflects a broader trend in the banking industry towards greater customization and personalized services*

Banks enhance SME services to tap into a growing market, foster loyalty, and stand out from competitors. Tailored solutions help manage risks, improve efficiency through digital innovations, and support economic growth, driving revenue and strengthening long-term client relationships.

Adoption challenges

Credit Risk: Limited credit histories and higher default rates among SMEs.	Tech Upgrades: Significant investment needed to update legacy systems.	Regulations: Complex regulatory environment and economic fluctuations.

Enablers

????	????	????

Key Use cases

1. Enhancement customer experience (Businesses) - onboarding/servicing
2. Leverage data & intelligence using OB e.g.working capital forecast
3. Value add to SME using ecosystem plays

Key Benefits

1. High margin business as opposed to Retail
2. SMEs tend to be sticky and more relational as they grow their business
3. Upsell opportunities through value add offerings

Industry examples

BOI/AIB - nCino modernisation of origination and servicing of SME lending

BBVA/Santander - Metro Bank pivoting from Retail to SME banking

01 Enhanced Human

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Customer Centric AI Driven Banking

Augmented Workforce

Augmented Workforce

- High personalise sevices
- Alway on enhanced convenienced
- Increased efficiencies through Automation
- Increaced Accuracy
- Better workforce utilisation
- Improce customer satisfaction
- Enhance risk management

Overview: *Banks are modernizing to stay competitive, focusing on both front-end and back-end operations. Manual processes are being replaced by AI, enhancing customer service, risk management, fraud detection, and process automation. Key technologies like intelligent document processing, predictive analytics, and autonomous intelligent agents are improving efficiency and time-to-market. A lean approach to banking operations is emerging, with some processes moving towards real-time scenarios. The concept of an augmented workforce is gaining traction, with generative AI transforming the traditional workforce into a more efficient, waste-eliminating "augmented" workforce.*

Adoption challenges

- Complex Integration with existing bank systems
- Maintaining the human touch
- Ensuring Responsible AI

Enablers

- Define AI and Automation blueprint
- Identify areas where AI and Automation will have biggest impact
- Implement blueprint and automate

Key Use cases

1. Enhancement business customer experience
2. Leverage data & intelligence
3. Value add to SME using ecosystem plays

Key Benefits

1. High margin business as opposed to Retail
2. SMEs tend to be sticky & more relational as they grow their business
3. Upsell opportunities & value add

Industry examples

Salesforce : Salesforce Agent Force: This example highlights the use of AI agents to deliver hyper-personalized services.

DBS Bank :DBS Bank utilizes AI agents to provide always on peronalised service to customers

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Customer Centric AI Driven Banking

Embedded/Subscription Finance

Embedded/Subscription Finance

- High personalise Services
- Value for Money
- Seamless digital Experience
- More Choice
- Integration with non banking eco systems

Overview: *Subscription-based is gradually entering the financial services industry, driving fintech partnerships and open banking initiatives. To convince customers to pay for services they previously considered free, banks must offer genuine value or cost savings. Key questions for banks include understanding customer needs and effectively delivering them. Inspiration can be drawn from fintechs like Revolut, NuBank, and N26, which offer benefits such as no ATM fees, cashback, and exclusive services. Subscription models can also add personalization through bundled services. Coupled with diversification through new revenue sources like superapps, vertical marketplaces, Banking-as-a-Service, and embedded finance supported by open banking.*

Adoption challenges

Convincing customers to pay for previously free services.	Managing high churn rates if customers don't see continuous value.	Ensuring secure storage of payment information to prevent data breaches.

Enablers

Scalable technology infra to deal with scale	Forming ecosystems and know your data (How do you leverage data)	Easy consumable marketplace model service

Key Use cases

1. Distribution of payment service across value chains at a customer level
2. Productions of Payment and banking service like credit at a customer level
3. Banking services for B2B

Key Benefits

1. Increase revenue at high margin reducing cost
2. Retain customer loyalty against Fintech
3. New channels to Market

Industry examples

Revolut: This bank has started a model that merges subscription-based banking with an embedded finance mindset.

HSBC : is working on embedded finance by implementing Vault (Thought Machine's product) smart contracts

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Intelligent & Responsible Banking services

"The future of banking is not just about algorithms and automation; it's about harnessing the power of AI to enhance, not replace, the human touch. Intelligent banking, at its core, means using technology responsibly to build stronger relationships with customers, understand their needs more deeply, and contribute to a more equitable and financially inclusive society."

Ana Botín

Executive Chair, Banco Santander

Intelligent & Responsible Banking: Underlying trends

01

Banking for All: The World Bank estimates that 1.4 billion people worldwide are unbanked. This trend seeks to drive for inclusion, against inequality, and for the simplification of access to financial services

02

Regulatory Compliance: Continued expansion of regulatory environment will be a focus for all banks as financial ecosystem adapt to an evolving world. Regtech will continue to play a virial role to navigate localization of EU and related legislation

03

Wealth: Bank Customers are seeking more sophisticated and personalized investment options to grow and safeguard their wealth. With the rise of digital platforms and data analytics, wealth managers have access to powerful tools that allow for more precise risk assessment, portfolio optimisation, and asset allocation

04

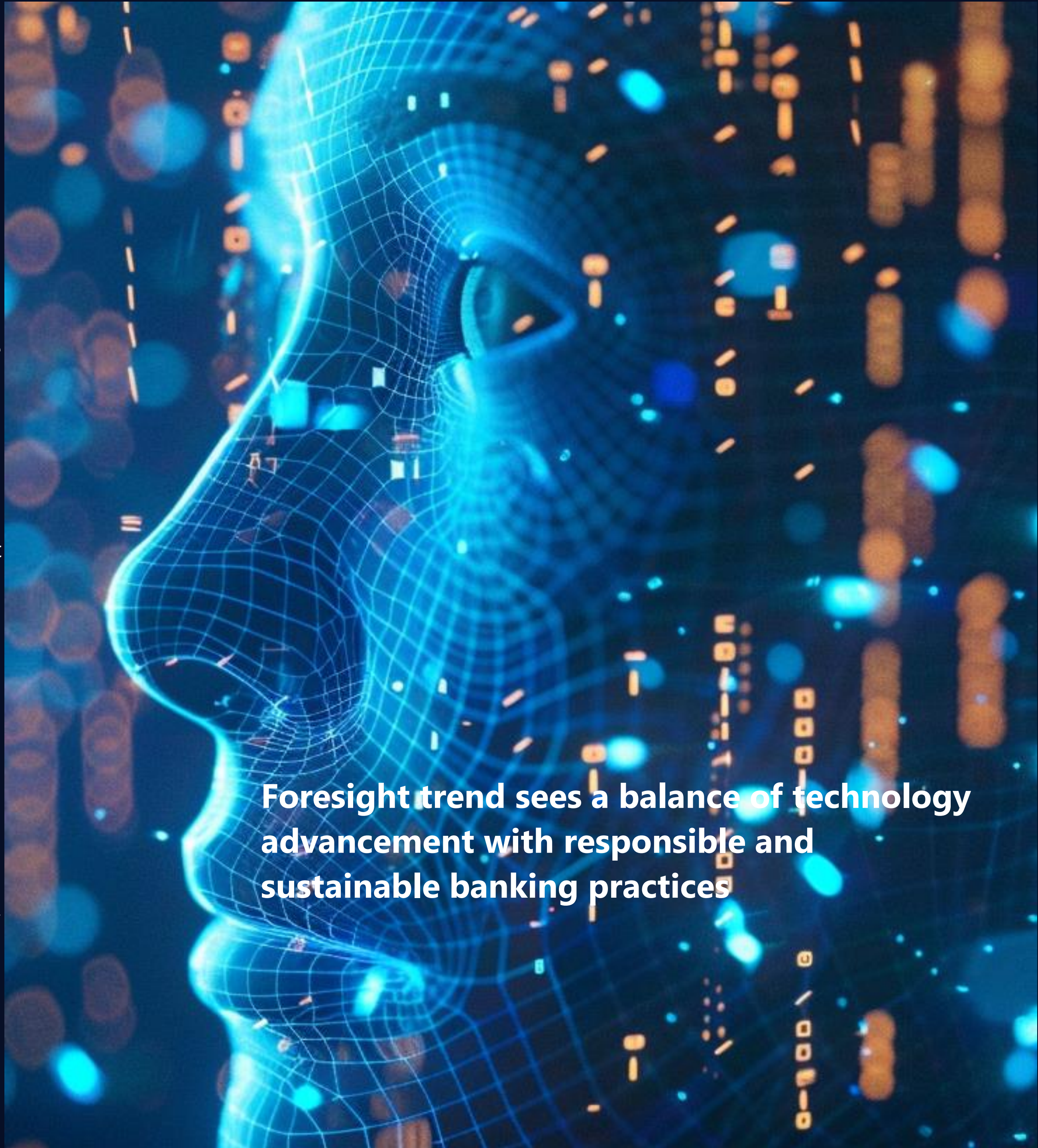
Payments (Cross Border/A2A): Real-time networks have made instant payments the standard for domestic transactions while global transfers have been slow to adapt. Outdated core systems in banks and complex international regulations continue to hinder global transfers

05

CBDC: Central Bank Digital Currencies (CBDCs) are emerging to complement or replace fiat money, driven by the need to compete with private monetary alternatives. They offer benefits like stricter controls, financial stability, efficient and secure payments, enhanced tracking, and quick implementation of monetary policies.

06

Sustainable finance: The banking sector is increasingly focusing on sustainable finance and green banking as part of their broader strategy to integrate environmental stewardship with economic growth. This trend is driven by the need to address climate change and meet regulatory requirements, as well as growing customer expectations for more sustainable financial products and services.



Intelligent & Responsible Banking : underlying concepts

Intelligent Efficiency

Banking for All
- Convenience
- Personalised
- Accessible
- Transparency
- Trust
- Secure

Regulatory Compliance
- Enhnaced Privancy
- Enhanced Security
- Improved Service Qulaity
- Transparency
- Trust
- Compliance
- Enhanced Risk management

Wealth
- Enhance products
- Enhance services
- Automated
- Always on servcie
- Convenience
- AI Driven Financial Advisory

Payments (Cross Border/A2A)
- Faster Payments
- Enhanced Security
- Lower Transaction Fees
- Enhanced conveniaence
- Real Time payments
- Enhance product offerings
- Customer satisfaction

CBDC: Central Bank Digital Currencies
- Personalised Digital Wallet
- Enhanced data Privacy
- Secure digital transactions and payments
- Virtual digital Assets managment
- Secure Payment and FX
- Enhance security and fraud detection
- Data ownership

Sustainable Finance
- ????
- Augmented software engineering
- Collaboration platforms
- Digital skills development
- Innovation culture
- Technology communities
- Talent retention

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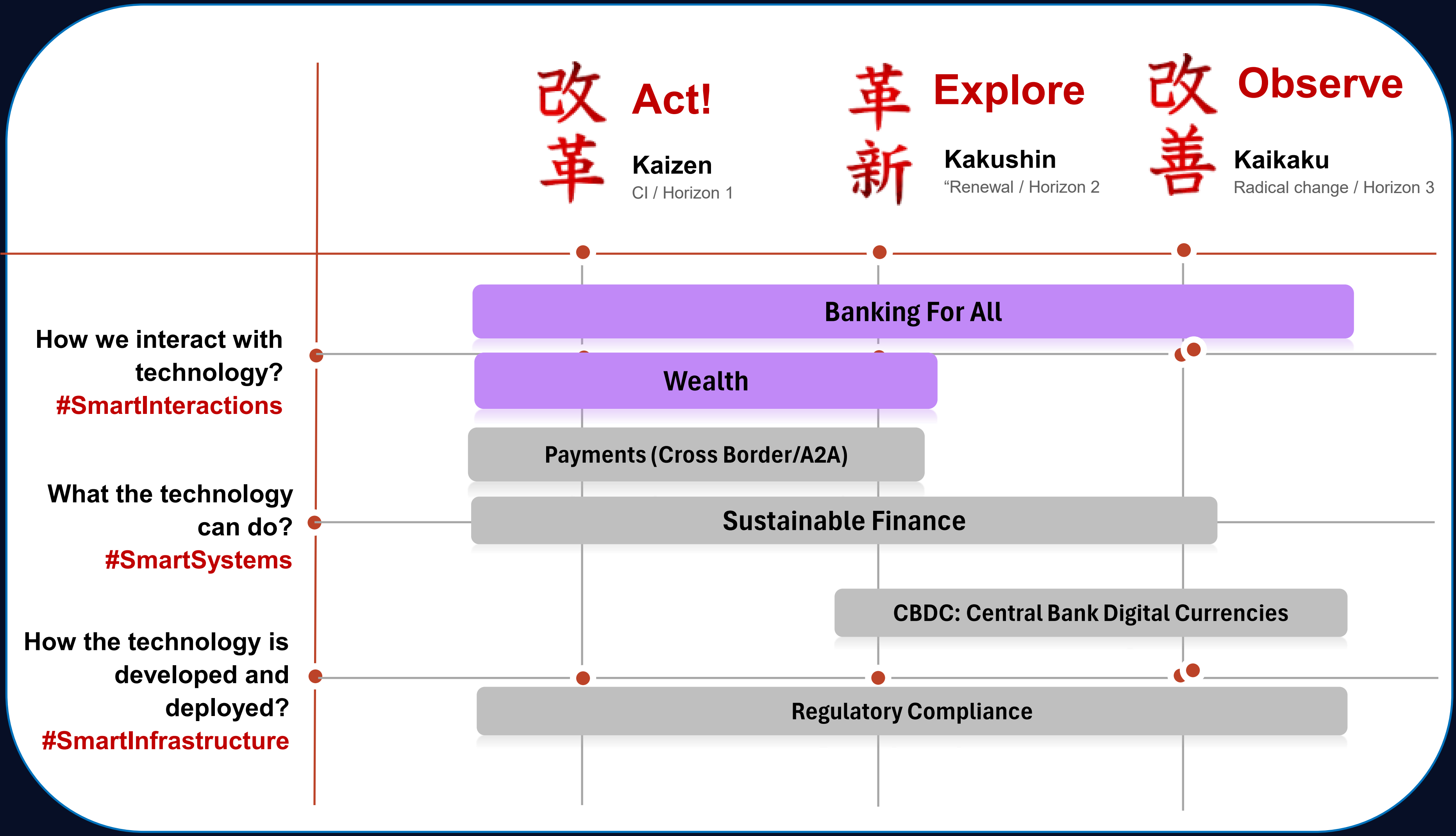
Foresight 2025 Technology Horizon : Customer Centric AI Driven Banking

Products and service for all customer full range customers

Enhance payment capabilities and increase service and flexibility

Changing role of physical currencies in context of evolving regulatory landscape



Intelligent & Responsible Banking

Banking for All

Banking for All

- Convenience
- Personalised
- Accessible
- Transparency
- Trust
- Secure

Overview: *Ensuring that everyone has access to banking services is a key priority for the sector, with a focus on inclusion, reducing inequality, and simplifying access to financial services. Barriers such as age, culture, or economic status should not prevent access. According to the World Bank, 1.4 billion people are currently unbanked. Efforts are being made to adapt products and services to meet the needs of disadvantaged groups and to simplify learning and training to encourage new financial habits.*

Generative AI can play a significant role by creating customized financial products, mitigating lending risks for those with limited credit history, assisting in financial education, and providing personalized guidance and advice. Banking for all will help banks address there ESG targets and improve their reputation but it is not without its challenges

Adoption challenges	Regulatory and mandate compliances	Developing technology that is accessible and cost effective as well as secure and easy to use.	Building trust and engaging with customer especially those that have been historically excluded from banking service.
Enablers	Customer experience to redesign engagement	Data and personalization services	High quality financial education content

Key Use cases

1. Inclusive and ethical finance
2. Fair, transparent lending practices
3. Financial well being (move away from products and services to improving financial well being)

Key Benefits

1. Impacts customer retention, churn
2. High value customers, lower CAC
3. Reduce regulatory penalties by aligning to the consumer duty vision

Industry examples

Grameen bank - (Bangladesh) Microfinancing,

Lloyds - Tailored support for vulnerable customers

02 Ambient Intelligent Experiences

03 Digital sustainability for economic resilience

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Intelligent & Responsible Banking

Regulatory Compliance

Regulatory Compliance

- Enhnaced Privancy
- Enhanced Security
- Improved Service Qulaity
- Transparency
- Trust
- Compliance
- Enhanced Risk management

Overview: *In the banking sector, trust, risk, and security are paramount, especially with the increasing use of Generative AI (Gen AI). Key regulations and frameworks have been established to address these concerns. The Digital Operational Resilience Act (DORA) mandates digital resilience for financial institutions in Europe, ensuring they can withstand and recover from all types of ICT-related disruptions. Similarly, the General Data Protection Regulation (GDPR) enhances data governance and privacy, ensuring that personal data is handled with the utmost care and security. As the banking sector continues to evolve with new technologies and increasing data growth, these regulations play a vital role in ensuring that financial institutions remain resilient, secure, and trustworthy. They help in combating fraud, protecting customer data, and maintaining trust in the financial system*

Adoption challenges	Data Management, security and sovereignty	Operational Resilience	New regulation (Green Finance, Cybersecurity enhancement, Open Banking, AI)
Enablers	Resilience Infrastructure and operations	Dynamic Cyber Security	Banking Modernisation

Key Use cases

1. BCBS 239
2. DORA/Op Res
3. MiFid 2, EMIR – Transaction reporting

Key Benefits

1. Maintain compliance posture
2. Strategic architecture approach could help drive efficiencies
3. Resilience/DORA gaps have an impact on CX driving switch for digital first banks

Industry examples

Mizuho: Regulatory Reporting EMIR Refit, MiFid 2

Santander UK: BCBS 239, FSCS regulatory reporting using data

ICBCS, Mizuho: DORA Cyber Risk

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Intelligent & Responsible Banking

Wealth

Wealth

- Enhance procusts
- Enhance services
- Automated
- Always on servcie
- Convenience
- AI Driven Financial Advisory

Overview: *Wealthy individuals are increasingly seeking sophisticated and personalized investment options to grow and protect their wealth. Digital platforms and data analytics provide wealth managers with powerful tools for precise risk assessment, portfolio optimization, and asset allocation. Significant growth in wealth management is expected in APAC, particularly in countries like India, China, Singapore, Hong Kong, and Japan. As market volatility and economic uncertainties continue, the demand for professional wealth management services is rising, highlighting the need for skilled financial advisors to navigate complex financial markets and deliver valuable insights to clients.*

A bank aims to deliver improved wealth services to attract and retain high-net-worth clients, increase assets under management, and boost revenue. Enhanced services, supported by digital tools and data analytics, enable precise risk assessment and personalized investment strategies, fostering stronger client relationships.

Adoption challenges	Technology Integration: Implementing advanced digital tools requires significant investment and expertise.	Client Trust: Building trust and educating clients about new digital tools can be difficult, especially for less tech-savvy individuals.	????
Enablers	Gen AI/CX for Wealth use case development	Monitor compliance for advice and flag challenges	Data governance transformation to drive analytics

Key Use cases

1. Enhance advisor and customer experience
2. Augment compliance in advice stage
3. Streamline investment operations including data transformation

Key Benefits

1. Enhanced revenue by freeing up IC time
2. Efficiency
3. Better data to drive Alpha via better portfolio strategy

Industry examples

Wealthify: Leveraging AI to drive digital investing

Blackrock: Using AI to drive investment research analysing transcripts and news as it emerges to assess the impact on its portfolio

02 Ambient Intelligent Experiences

03 Digital sustainability for economic resilience

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Intelligent & Responsible Banking

Payments (Cross Border/A2A)

Payments (Cross Border/A2A)

- Faster Processing Times
- Enhanced Security
- Lower Transaction Fees
- Enhance conveniaence
- Real Time payments
- Enhance product offerings
- Customer satidfaction

Overview: *The increasing flow of remittances, the deepening of global supply chains, and the expansion of online merchants into new markets are driving the need for modernization in cross-border payments. While real-time networks have made instant payments standard for domestic transactions, global transfers remain slow due to outdated core systems in banks and complex international regulations. These transfers can take several days and cost up to ten times more than domestic ones.*

The shift from traditional methods to API connectivity, along with regulatory changes like PSD3, PSR, and ISO 20022, is beginning to address these issues. Exciting use cases, including smart contracts and real-time liquidity, are also emerging.

Banks aim to improve cross-border payments to offer faster, cheaper, and more reliable services, enhancing customer satisfaction and staying competitive. This modernization also reduces costs, supports global trade, and meets regulatory standards.

Adoption challenges

- Navigating complex and evolving regulations across different countries
- Protecting against increasingly sophisticated cyberattacks
- Integrating new technologies with legacy systems can be costly and time-consuming

Enablers

- Clear Key Benefits development
- Monitor compliance for advice and flag challenges
- Data governance transformation to drive analytics

Key Use cases

1. Real Time payments
2. Cross border payments
3. Open Banking adoption, NPA/NPV

Key Benefits

1. High competition from FinTechs
2. Monetisation opportunities using commercial relationships
3. Wholesale flows growing in multi currency formats as more countries move away from dollar

Industry examples

Ripple: Ripple's technology enables real-time, cross-border payments with end-to-end tracking and certainty leveraging blockchain

<SEPA SCT: new SEPA regulatory changes include the Instant Payment Regulation (Regulation 2024/886), which is designed to unlock the full potential of instant payments across Europe

Intelligent & Responsible Banking

CBDC: Central Bank Digital Currencies

Payments (Cross Border/A2A)

- Faster Processing Times
- Enhanced Security
- Lower Transaction Fees
- Enhance conveniaence
- Real Time payments
- Enhance product offerings
- Customer satidfaction

Overview: *Central Bank Digital Currencies (CBDCs) are being explored as potential alternatives to traditional fiat money, aiming to counter private monetary options. They offer advantages like improved controls, financial stability, and payment efficiency. Currently, 134 countries, covering 98% of global GDP, are investigating CBDCs, with 66 in advanced development stages. All G20 nations are involved, and the BIS's Agora project includes central banks and major financial institutions focusing on tokenizing cross-border payments.*

The adoption of digital currencies faces several challenges. These include navigating complex regulations, ensuring cybersecurity, and integrating new technologies with legacy systems. Additionally, there are concerns about data privacy, consumer trust, and the need for significant investment in technology upgrades

Adoption challenges

Navigating complex and evolving regulations across different countries	Protecting against increasingly sophisticated cyberattacks	Integrating new technologies with legacy systems can be costly and time-consuming

Enablers

Clear value proposition development	Monitor compliance for advice and flag challenges	Data governance transformation to drive analytics

Key Use cases

1. Cross Boarder payments
2. Embedded finance – Integrating DC into everyday applications and services
3. Enhance privacy, security and Fraud protection

Key Benefits

1. Enhance efficiency and speed – Reducing the cost to serve
2. Reduce transition cost removing need for intermediaries
3. Improve customer experience and Loyality

Industry examples

Bank of England–Ainvolved in the exploration of Central Bank Digital Currencies (CBDCs)

European Central Bank (ECB):– Working with national central banks of the euro area to look into the possible issuance of a digital euro.

The bank of France– They have been testing the use of a digital euro for wholesale transactions and cross-border payments

02 Ambient Intelligent Experiences

03 Digital sustainability for economic resilience

05 Accelerated security fusion

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Intelligent & Responsible Banking

Sustainable finance (Green Banking)

Payments (Cross Border/A2A)

- Enhance ESG portfolio
- Customer satisfaction
- Customer Loyalty
- Diverse product and services
- Regulatory
- Reduce carbon footprint
- Customer satidfaction

Overview: *The banking sector is increasingly focusing on sustainable finance and green banking as part of their broader strategy to integrate environmental stewardship with economic growth. This trend is driven by the need to address climate change and meet regulatory requirements, as well as growing customer expectations for more sustainable financial products and services1.*

Banks worldwide are turning to sustainable initiatives to save their reputation and meet the Sustainable Development Goals (SDGs) set by the United Nations. Various providers have come up with standardized scores to measure and rank companies according to environmental, social, and governance (ESG) criteria4. In a recent U.S.-based Visa survey, 72 percent of respondents indicated that sustainability was very important to them, and 85 percent of global consumers have altered their purchasing behaviors in the last five years to become more sustainable.

Adoption challenges

Banks face the challenge of navigating a complex and evolving regulatory landscape.	Obtaining reliable and high-quality ESG-related data is a significant barrier.	Banks often face pressure to deliver strong short-term financial results, which can conflict with long-term ESG goals

Enablers

The adoption of open banking continues to grow, creating an ecosystem with fintechs and other FS providers	Develop Green Finance strategy with integration to Customer feedback and ESGs	Leverage Data and analytics to build green products and services

Key Use cases

1. Green bank Accounts
2. Climate credit card and Lotalty programs
3. ESG investing platforms

Key Benefits

1. Manage risks associated with climate change
2. Customer Loyalty and TrustI
3. Regulatory Compliance and Competitive Advantage

Industry examples

International Finance Corporation (IFC)–eading provider of green loans

World Bank: Since 2008, the World Bank has issued over USD 20 billion equivalent in Green

BNP Paribas : have issued green bonds to finance projects with positive environmental impacts(ie. renewable energy and energy efficiency projects)

Digital Banking Modernisation

"Digital banking modernization isn't about simply layering new technology onto old systems. It's about fundamentally rethinking how we serve customers in a digital-first world, creating seamless, personalized, and intuitive experiences that empower them financially, while ensuring that no one is left behind in the digital transition."

Cathy Bessant

Former Chief Operations and Technology Officer, Bank of America

Digital Banking Modernisation : underlying trends

1. **Banking Modernisation :** Banks are increasingly focused on modernization and cloud adoption to stay competitive, impacting both front-end and back-end operations. Tailored cloud environments, primarily using hybrid, multi-cloud, and poly-cloud configurations, are preferred, while mono-cloud approaches are generally avoided for systemic reasons.

2. **Operational Resilience:** Operational resilience has become a critical trend in the banking sector, emphasizing the importance of maintaining continuous service delivery amidst disruptions. Banks are increasingly focusing on identifying and mitigating potential risks, developing robust business continuity plans, and establishing effective incident response protocols.

3. **Adaptive Security:** enables organizations to continuously monitor, detect, and respond to threats. With AI's advent, this imperative grows increasingly complex amid the ongoing cyber warfare arms race, compelling banks to develop comprehensive, dynamic security strategies.

4. **Eco-AI and Cloud:** banks face the ongoing challenge of leveraging AI benefits, while transitioning to Green-Ops practices and sustainable-by-design approaches to lead the charge toward net-zero and negative carbon emissions.

Digital Banking Modernisation: underlying concepts

Customer Centric AI Driven Banking

Digital Banking Modernisation

- Streamlined platforms and Services
- Customer satisfaction
- Staff Satisfaction
- Customer retention
- Remove technical Debt
- Future proof IT services
- Remove legacy constraints

Operational Resilience

- 24/7 Availablity
- Enhanced Security
- Customer Trust
- Proactive Monitoring
- Self Healing Infrastructure
- Future proof IT services
- Remove legacy constraints

Adaptive Security

- Adaptive Authentication & AI managed Access
- Behavioural AI analytics
- Continuous & multi-layered authentication
- Dynamic & multi- agentic threat response
- Hyper-automated SecOps
- Next-gen encryption
- Real-time resilience
- Zero Trust everything

Eco AI & Cloud

- AI enhanced energy optimisation
- Agentic CCoE
- Data virtualisation & Zero ETL
- GreenOps-FinOps convergence
- Low-carbon LLMs
- Low-Energy by Design
- Private LLMs (Gov-GPT)
- Quantization & Model size reduction

These are more customer expectation / value statements and not underlying concepts. Review to see if they make sense.

Perhapse ad a short description to contextualise

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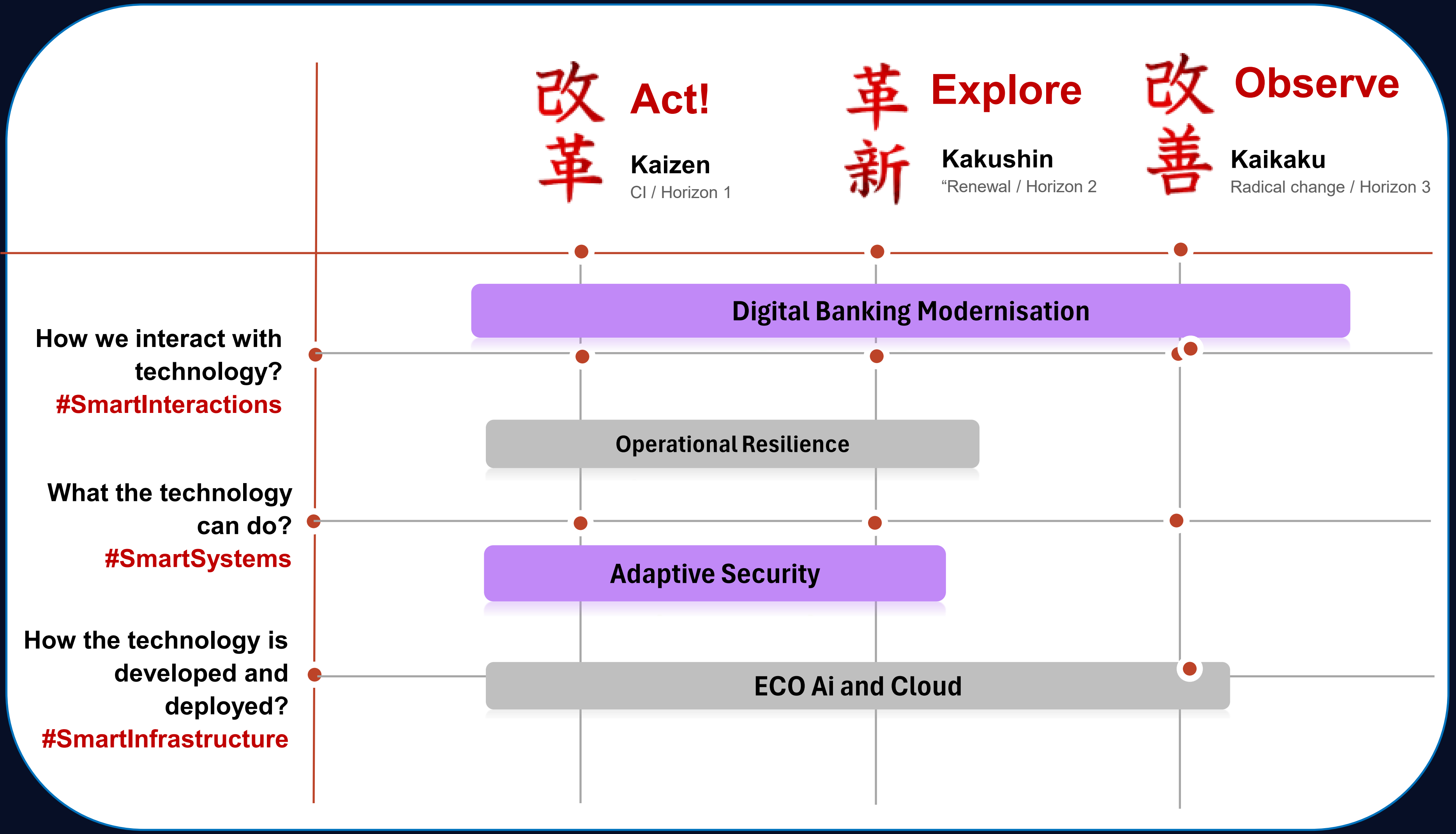
Foresight 2025 Technology Horizon : Customer Centric AI Driven Banking

Ongoing initiative to Modernise Legacy across all Horizons

Regulation driving Operational resilience for business and technology include enhance security posture

Leveraging AI and the cloud to meet ESG target and balance green IT with increase demand for processing through AI



Digital Banking Modernisation

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Overview: *Banks are increasingly focused on modernization and cloud adoption to stay competitive, impacting both front-end and back-end operations. However, mainframe environments pose challenges due to high maintenance costs, integration difficulties, scalability issues, security vulnerabilities, and innovation obstacles. Transformation pathways include business-driven, IT-driven, and mainframe-driven approaches. Tailored cloud environments, primarily using hybrid, multi-cloud, and poly-cloud configurations, are preferred, while mono-cloud approaches are generally avoided for systemic reasons.*

Modernizing banking is a crucial priority for the industry, motivated by the necessity to remain competitive and improve both front-end and back-end processes. However, this transition presents several challenges, especially in managing legacy systems while incorporating new technologies. The integration of new technologies with existing legacy systems can be both expensive and time-consuming

Adoption challenges	High Maintenance cost for maintaining legacy systems	Integration challenges	Scalability issues
Enablers	Develop customer journeys and system map to drive modernization activities	Leverage AI transformation technology to mitigate transformation risk	Develop target stare for modernisation journey

Key Use cases

1. User centric and data driven approach to modernization
2. Address Operational Resilience for Regulation requirements
3. Define cloud journey to enable business

Key Benefits

1. Moderisation to meet customer expectations
2. Keep ahead or regulation, building a stronger enterprise
3. Enable a date driven enterprise and scalable business

Industry examples

Bank of Australia: Overhaul of core banking platforms

BBVA: Early adopter of Cloud and API driven platforms

ING: "Think Forward" Strategy with emphasis on digital and modernisation

Digital Banking Modernisation

Operational Resilience

Operational Resilience

- 24/7 Availablity
- Enhanced Security
- Customer Trust
- Proactive Monitoring
- Self Healing Infrastructure
- Future proof IT services
- Remove legacy constraints

Overview: Operational resilience has become a critical trend in the banking sector, drive by the introduction of DORA act in the EU, but also by common sense for the industry, emphasizing the importance of maintaining continuous service delivery amidst disruptions. Banks are increasingly focusing on identifying and mitigating potential risks, developing robust business continuity plans, and establishing effective incident response protocols. The goal is to ensure that critical functions can continue during and after disruptions, such as cyber-attacks, natural disasters, and system failures. Additionally, banks are implementing recovery strategies to restore normal operations swiftly and adhering to regulatory guidelines to enhance their operational resilience. This trend underscores the banking industry's commitment to stability, reliability, and customer trust in an ever-evolving landscape.

Adoption challenges

- Legacy systems and Infrastructure
- Complex Regulatory compliance landscape
- Cyber security Threats

Enablers

- Cloud strategies for resilient Infrastructure
- Business recovery and BCP strategies
- Telemetry evolving to Observability strategies

Key Use cases

1. Always on production
2. Cyber incident response and recovery
3. Continuous monitoring and proactive maintenance

Key Benefits

1. 24/7 resilient operation with minimal down time
2. Meet regulatory requirement and customer expectation for business resilience
3. Proactive management reduce maintenance costs

Industry examples

Financial Conduct Authority: FCA regulations require firms to ensure they can continue to provide critical services during disruptions or face serious consequences

DORA: Non-compliance with DORA can lead to substantial fines and penalties as well as significant reputational damage

03 Digital sustainability for economic resilience

05 Accelerated security fusion

Digital Banking Modernisation

ADAPTIVE SECURITY

Adaptive Security

- Adaptive Authentication & AI managed Access
- Behavioural AI analytics
- Continuous & multi-layered authentication
- Dynamic & multi- agentic threat response
- Hyper-automated SecOps
- Next-gen encryption
- Real-time resilience
- Zero Trust everything

Overview: *The democratization of AI has enabled individuals to pose significant cybersecurity threats. Security departments must automate routine tasks to focus on strategic solutions. Behavioural AI can prevent sophisticated attacks, like the 2020 Twitter hack. Dynamic, multi-agent solutions will soon allow near-instantaneous threat response. Hyper-automation in SecOps (e.g., SIEM, SOAR, TIP) will reduce human effort and risk. As zero-trust principles become more prevalent, AI-managed authentication and authorization solutions will be increasingly adopted, alongside adaptive, real-time, continuous, and multi-layered authentication methods.*

Quantum computing threatens traditional encryption. PSOs can enhance security with blockchain, homomorphic encryption, and Zero-Knowledge Proof (ZKP) solutions. Centralized collaboration across government cyber defence communities is crucial.

Adoption challenges	Legacy systems integration	Budgetary constraints	Skills and attrition of deep Security Leadership	Maintaining compliance with regulation
Enablers	AI playbook and executive support for automation in cyber defence	AI will release capacity and reduce human for routine tasks	Cloud based supplier tools are mature and consumption based	Mature SecOps teams & established frameworks (Secure-by-Design)

Key Use cases

1. Dynamic Real time response to Threats
2. AI Automation for SecOps
3. Continuous & Multi Layered Authentication

Key Benefits

1. Enhanced Risk Management: detect and respond to threat in real time.
2. Improved Customer Trust
3. Enhance operational efficiency

Industry examples

Global Bank Corp: Established a next-generation Security Operations Center (SOC)

HSBC: HSBC has implemented adaptive security measures, including AI-driven threat detection and continuous monitoring,

Digital Banking Modernisation

ECO AI & CLOUD

Eco AI & Cloud

- AI enhanced energy optimisation
- Agentic CCoE
- Data virtualisation & Zero ETL
- GreenOps-FinOps convergence
- Optimised processing and data management
- Low-Energy by Design
- ESG Target
- Quantization & Model size reduction

Overview: *In the coming years, banks face converging technologies that will both reduce and increase their carbon footprint. Now that cloud is well established across the industry, teams will need to shift their attention to the end-to-end aspects of energy consumption and cost optimisation. As AI becomes mainstream,banks will need to exploit the power of AI to drive a reduction in carbon and cost. Paradoxically, meanwhile, the proliferation and demand for power-hungry infrastructure to support AI and LLMs means optimising end-to-end footprint for AI supported solutions will become critical, leading toward a low-energy-by-design mandate. Furthermore, we will witness the convergence of GreenOps and FinOps functions.*

Adoption challenges	Legacy systems integration	Budgetary constraints	AI readiness	Regulatory compliance
Enablers	Motivation and will to act to get to net-zero and drive AI Adoption	Clear action plans (AI Opportunities, Clean Power strategy)	Wide adoption of cloud and established CCoE/FinOps	AI and natural language technologies drive easier adoption despite skills shortages

Key Use cases

1. AI-Enhanced Energy OptimizationAI Automation for SecOps
2. GreenOps-FinOps Convergence
3. Low-Carbon Large Language Models (LLMs)

Key Benefits

1. Reduced cost of IT Operations
2. Balance the AI adoption paradox through Low carbon LLMs
3. Achieve corporate ESG targets

Industry examples

BBVA: BBVA has implemented Salesforce NetZero to support their sustainability initiatives

HSBC: HSBC has been leveraging Google Cloud technology to innovate and meet their climate commitments.

INPUTS AND WORKING MATERIAL

Tech Radar - Representing mainstream adoption

Analyzing technology trends and considering their evolution serves as your guide to the future, allowing you to foresee changes and prepare yourself for upcoming shifts. In the constantly changing tech landscape, keeping up with the newest advancements is essential, not just advantageous.